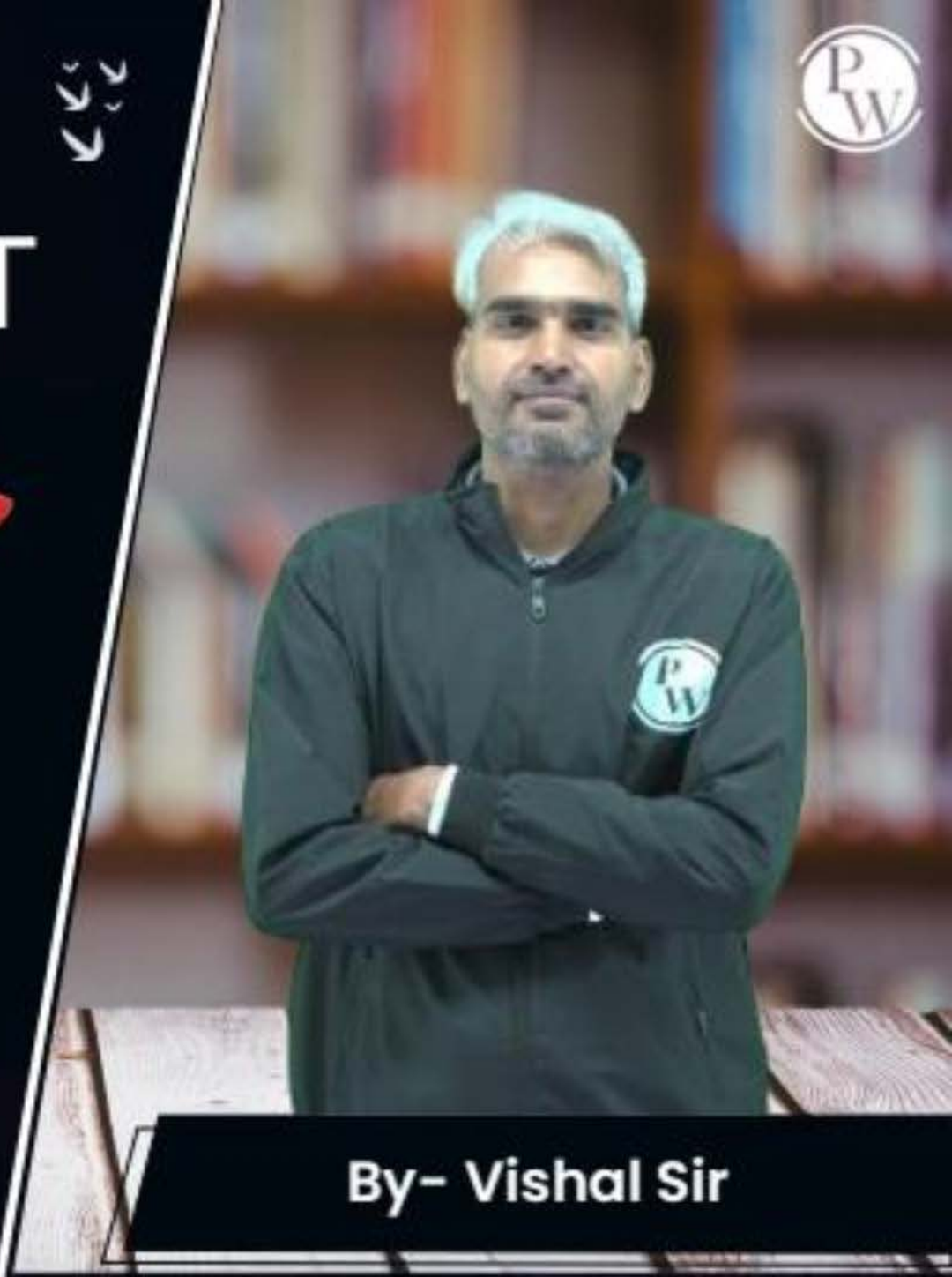
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Classification based on serializability
- **Topic** View serializable schedule
  1. Read from initial DB
  2. Write - Read Sequence
  3. Final update
- **Topic** View equivalence condition

Slide

Topics to be Covered



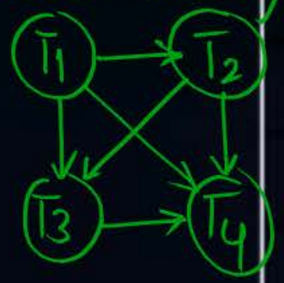
- **Topic** Concurrency control protocols
- **Topic** Simple use of shared and exclusive locks
- **Topic** Two phase lock (2PL)

#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

w.r.t Conflict serializability



Topological order
$= T_1 \to T_2 \to T_3 \to T_4$

Given schedule is C.S.S. & Conflict equivalent serial schedule is $T_1 \to T_2 \to T_3 \to T_4$

① Read from initial DB:
A: No write of A ∴ No Constraint
B: No read from initial DB ∴ No Constraint

② WR Sequence:
A: No WR seq
B: No WR seq
} No Constraint

③ Final update
A: No Constraint
B: $(T_1, T_2, T_3) \to T_4$
$3! = 6$ ways

∴ Given schedule is view equivalent to 6 serial schedules

Q: How many serial schedules are there which are view equivalent to given schedule, but not conflict equivalent to given schedule?

Ans = # View equivalent serial schedules − # Conflict equivalent serial schedules
$= 6 - 1 = 5$

#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3
R(A)		
	R(B)	
	W(B)	
		R(A)
		W(A)
R(B)		
		R(B)
	R(A)	
	W(A)	

① A: $T_1 \rightarrow (T_2, T_3)$ & $T_3 \rightarrow T_2$

B: No Constraint

② A: $T_3 \rightarrow T_2$

B: $T_2 \rightarrow (T_1, T_3)$ ∴ $T_2 \rightarrow T_3$

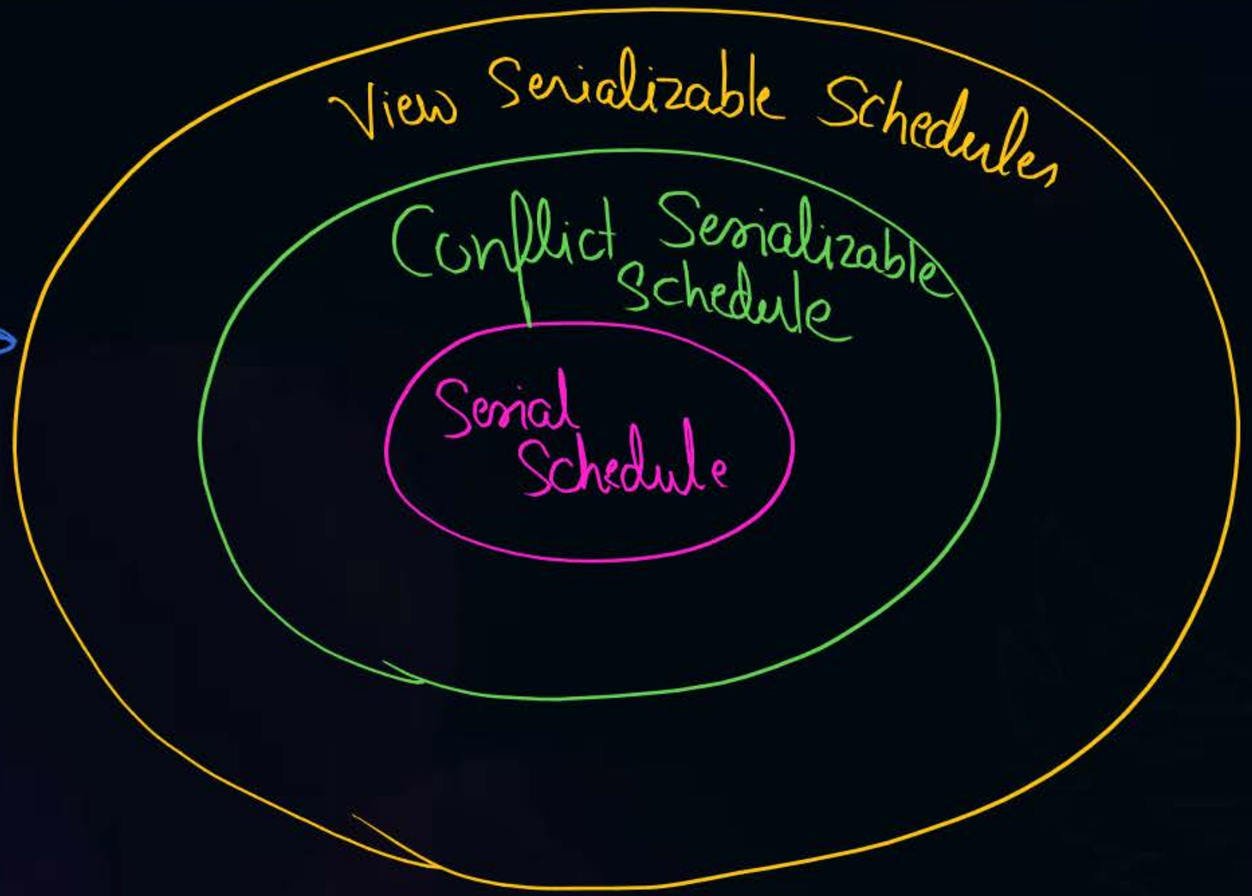
Cycle ∴ Not a V.S.S.

③ A: $T_3 \rightarrow T_2$

B: No Constraint



Topic : Conflict serializable & View serializable schedule



Overall Serializable Schedules



Topic : Concurrency Control Protocols

- Concurrency Control Protocols are responsible for avoiding the execution of non-serializable schedules.

- Concurrency Control Protocols are suggested to ensure serializability



Topic : Concurrency Control Protocols

- There are two types of concurrency control protocols.

1. Lock based Protocols
   - Two Phase Locking Protocol (2PL)
     - (i) Basic 2PL
     - (ii) Strick 2PL
     - (iii) Conservative 2PL
     - (iv) Rigorous 2PL
2. Time-stamp based Protocols
   - (i) Basic Time stamp ordering Protocol.
   - (ii) Thomas write Time-stamp ordering Protocol



Topic : Lock Based Protocols

Lock is a variable associated with a data item that describes a status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. Shared LOCK [S]: It is read only lock.

   A transaction will request for a Shared lock on a data item if transaction wants to perform only read oph on that dataitem

2. Exclusive LOCK [X]: It is read as well as write lock.

eg: Transaction
R(A)

it will be denied if dataitem is not locked appropriately

=> ie. Transaction
S(A) ← let (Read only) granted
R(A) ✓ Allowed
~~W(A)~~ denied
~~R(B)~~ denied
data item B is not locked by transaction



Topic : Lock Based Protocols

Lock is a variable associated with a data item that describes a status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. Shared LOCK [S]: It is read only lock.

eg: Transaction
R(A)
~~W(A)~~ denied

=> ie Transaction
X(A) - granted
W(A) ✓ Allowed
~~W(B)~~ denied
~~R(B)~~ denied
R(A) ✓ Allowed

If transaction wants to perform Read opn only then it is not allowed to request for Exclusive lock

2. Exclusive LOCK [X]: It is read as well as write lock.

If transaction wants to perform Write opn on a dataitem or wants to perform both read as well as write oply on a data item then it will request for Exclusive lock



Topic : Lock Compatibility Table

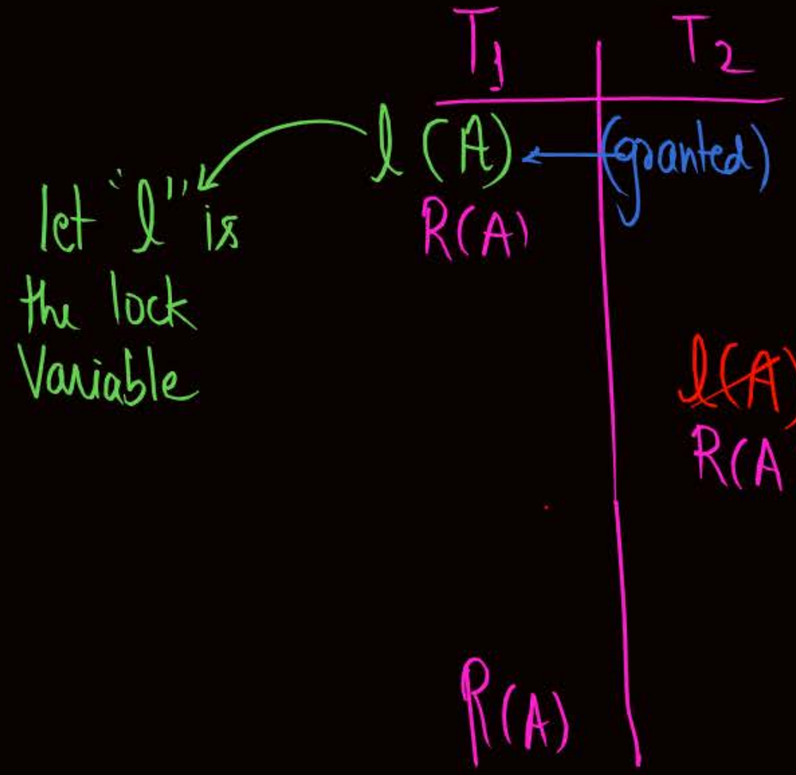
lock held by transaction T_i

data item = A	S	X
S	Allowed ✓	Not allowed ✗
X	Not allowed ✗	Not allowed ✗

lock requested by transaction T_j

Note: If we use single lock for both read and write operation then readers of the data item also will not be allowed to execute Concurrently

Eg: Consider the following schedule



Even readers of the same dataitem are not allowed to execute Concurrently

i.e. We need different lock for read & write oph, Here we use Shared & Exclusive locks

Note: To ensure serializability a non-serializable schedule must not be allowed to execute.

IMP



Topic : Simple use of shared and exclusive locks

i.e. No restriction in which dataitems Can be locked or unlocked.

Consider the following schedule

S

T_1	T_2
X(A)	
R(A)	
W(A)	
Unlock → U(A)	
	S(A)
	R(A)
	S(B)
	R(B)
	U(A)
	U(B)
X(B)	
R(B)	
W(B)	
U(B)	

→ Given schedule is a non-serializable schedule

↓

and

↓

it is allowed to execute using shared and exclusive lock, if there is no restriction in which dataitems Can be locked or unlocked

A non-serializable schedule is allowed to execute,

∴ Simple use of Shared & Exclusive locks does not ensure serializability



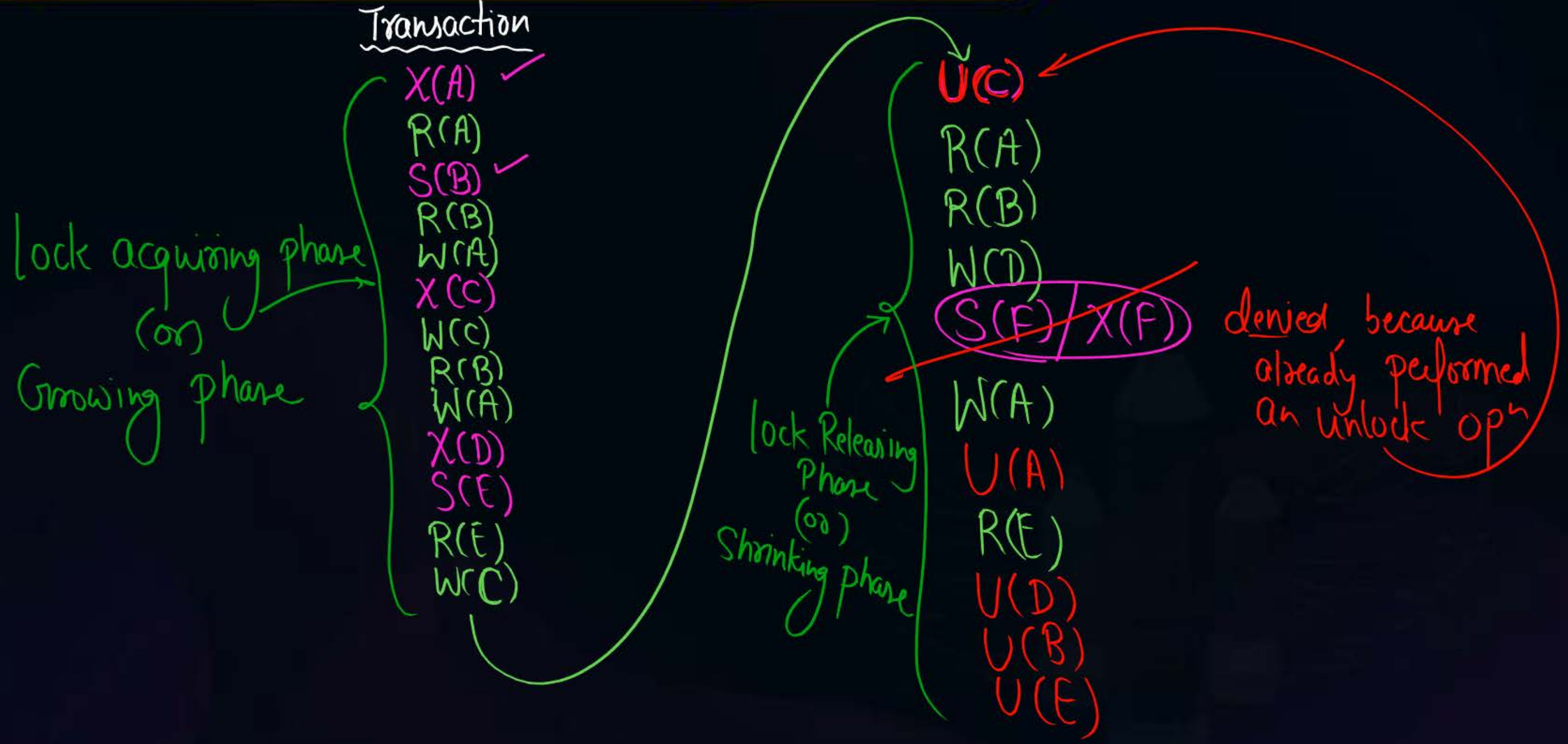
Topic : Basic Two Phase Locking Protocol

There are two phases in Two phase locking protocol

1. Growing phase / Lock acquiring phase
2. Shrinking phase / Lock releasing phase

In Two Phase Locking Protocol, transaction T is allowed to request for a lock only if transaction T has not performed any unlock operation.

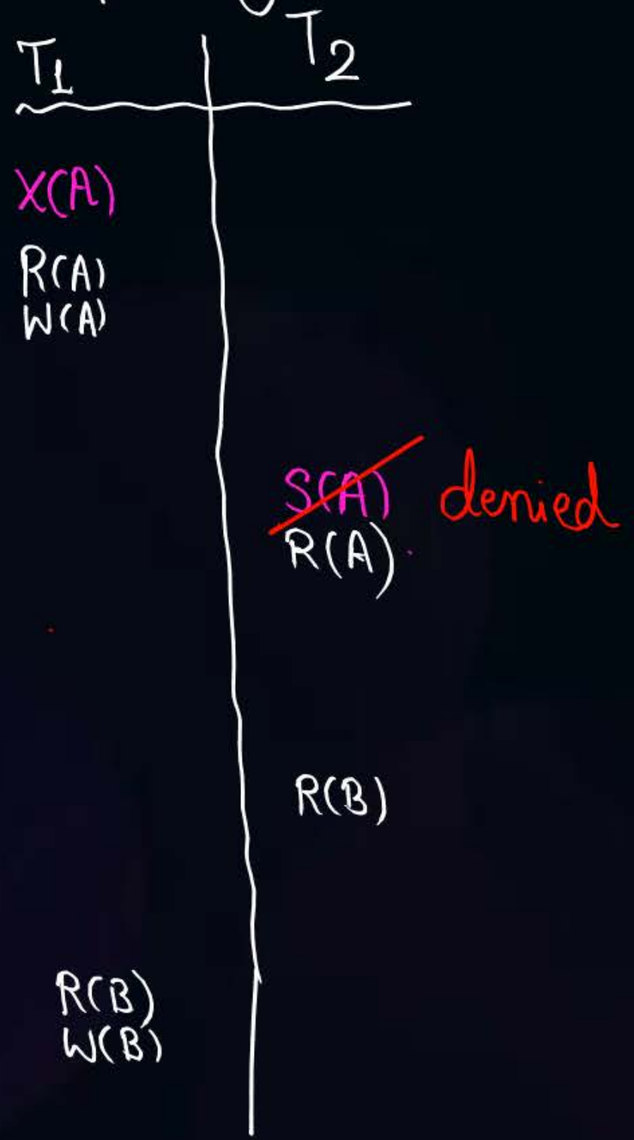
Topic : Basic Two Phase Locking Protocol





Topic : Basic Two Phase Locking Protocol

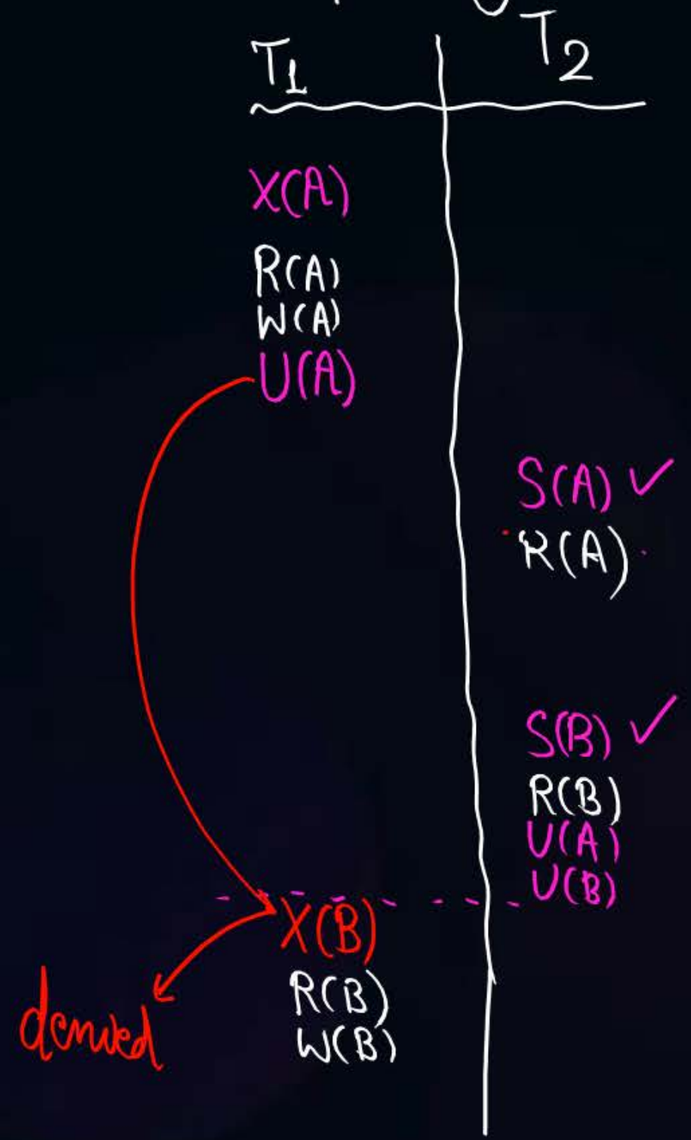
Consider the following schedule





Topic : Basic Two Phase Locking Protocol

Consider the following schedule





Topic : Basic Two Phase Locking Protocol

Consider the following schedule





Topic : Basic Two Phase Locking Protocol

Consider the following schedule



It is not serializable schedule
(and)
Not allowed to execute using Basic Two phase locking protocol

Basic 2PL



Topic : Basic Two Phase Locking Protocol

Consider the following Schedule

T_1	T_2
X(A)	
R(A)	
W(A)	
X(B)	
U(A)	
	S(A)
	R(A)
R(B)	
W(B)	
U(B)	
	S(B)
	R(B)
	U(A)
	U(B)

Precedence graph w.r.t Conflict serializability



Acyclic,
∴ Schedule is C.S.S.
And Conflict equivalent Serial schedule is $(T_1 \rightarrow T_2)$

Given schedule is a Conflict Serializable Schedule and allowed to execute using Basic 2PL



Topic : Basic Two Phase Locking Protocol

Consider the following schedule

T_1	T_2
	X(B)
	R(B)
S(A)	
R(A)	
~~S(B)~~ denied	
R(B)	
	W(B)

Precedence graph w.r.t. Conflict Serializability



Acyclic

∴ C.S.S.

&

Conflict equivalent serial schedule is

$T_1 \rightarrow T_2$

Given schedule is a Conflict Serializable schedule but Not allowed using Basic 2PL

Topic : Basic Two Phase Locking Protocol

Consider the following schedule

T_1	T_2
X(A) ✓	
R(A)	
	~~X(A)~~ denied
	W(A)
W(A)	
	W(A)

Given schedule is not a C.S.S, but it is a View serializable schedule (V.S.S) and View equivalent serial schedule is $(T_1 \rightarrow T_2)$

Schedule is not a C.S.S. but it is a V.S.S. i.e. it is a Serializable schedule but not allowed to execute using Basic 2PL



Topic : Basic Two Phase Locking Protocol

1. 2PL ensures serializability, i.e., a non-serializable schedule is never allowed to execute using 2PL.
2. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule, but every conflict serializable schedule need not be allowed to execute by 2PL. {i.e. There are some C.S.S. which are not allowed by 2PL}
3. If schedule is not a conflict serializable schedule, the that schedule is not allowed to execute using 2PL.
4. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule and conflict equivalent serial schedule can be given by order of lock points.

Will be discussed in the next class.



2 mins Summary

Topic Concurrency control protocols

Topic Simple use of shared and exclusive locks

Topic Two phase lock (2PL)

THANK - YOU